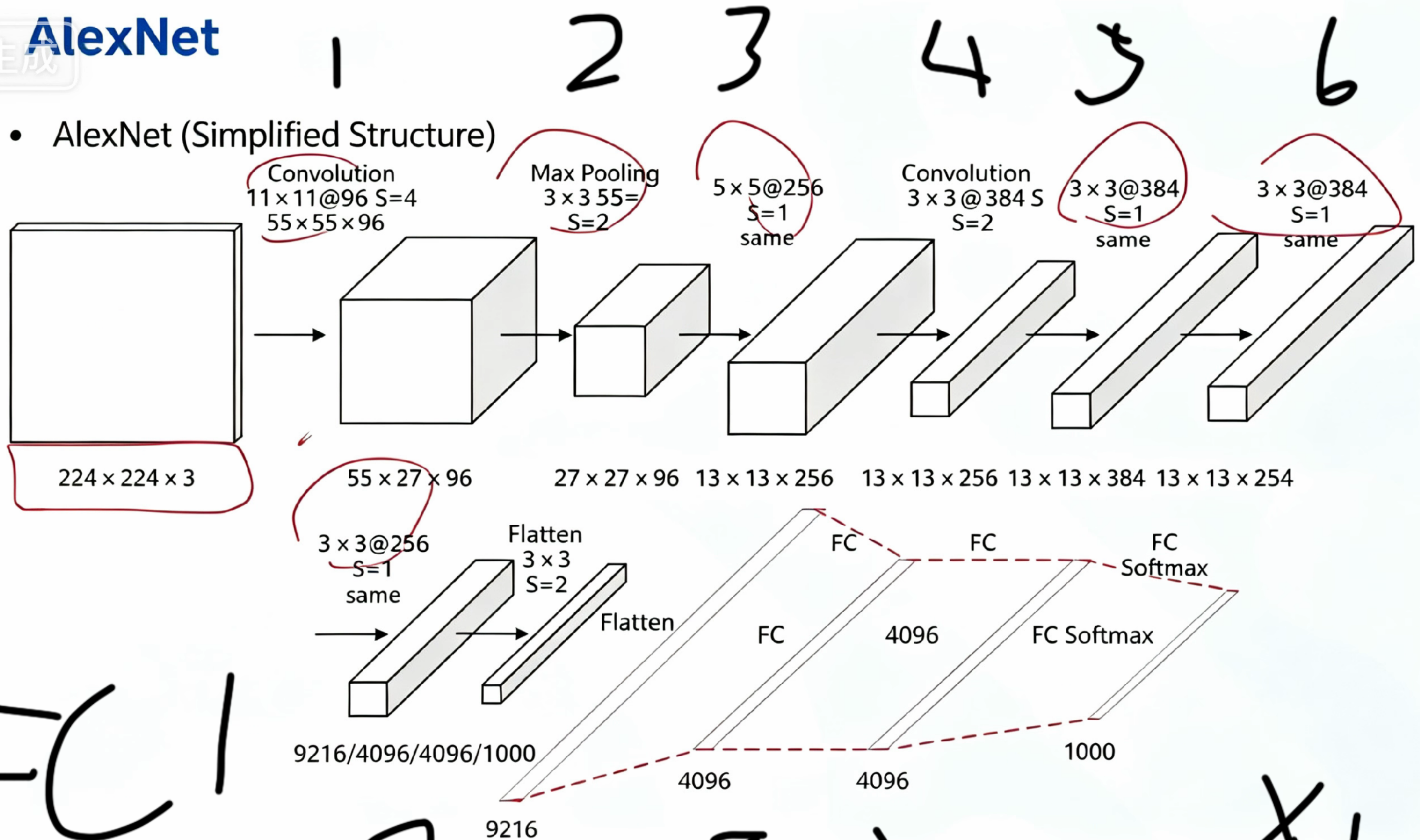
AlexNet

- AlexNet (Simplified Structure)



1: convoluntional layer includes 96 11 * 11 kernals with stride = 4-> output 55 * 55 * 96. 96 outputs divided into 2 groups, with 48 in each group.
2. maxing pooling with 3 * 3 kernal and stride = 2-> 27*27*96. Each group has 48 outputs.
3.96 11*11 kernal stride = 4 5*5 kernal stride = 1 -> 27*27*256
4. maxing pool 3*3 stride = 2 -> 13*13*256
5. convoluntional layer 256 3*3 stride =1 kernal, then max pooling 3*3 stride-=2 kernal-> after flatten, it ill have 9216 units.

FC1

8

weights

$$\begin{bmatrix} y_1 \\ \vdots \\ y_{4096} \end{bmatrix} = b + \begin{array}{ll} x_1 & [w_{1,1}, w_{1,2}, \ldots w_{1,9216}] \\ x_2 & [w_{2,1}, w_{2,2} \ldots w_{2,9216}] \\ \vdots & \ldots \\ x_{9216} & [w_{4096,} \ldots w_{4096,9216}] \end{array}$$

ReLU = max(0, x)

Dropout

Randomly select 50% of the neurons and force their outputs to be set to 0, so that they do not participate in the subsequent calculations.

FC2 weight(same)

ReLU → Dropout → FC3 → 1000 value

weight

1000×4096

softmax

$$p(\text{category} = k) = \frac{e^{z_k}}{\sum_{j=1}^{1000} e^{z_j}}$$